



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID BP Physics

a.c. generator

Class Outline

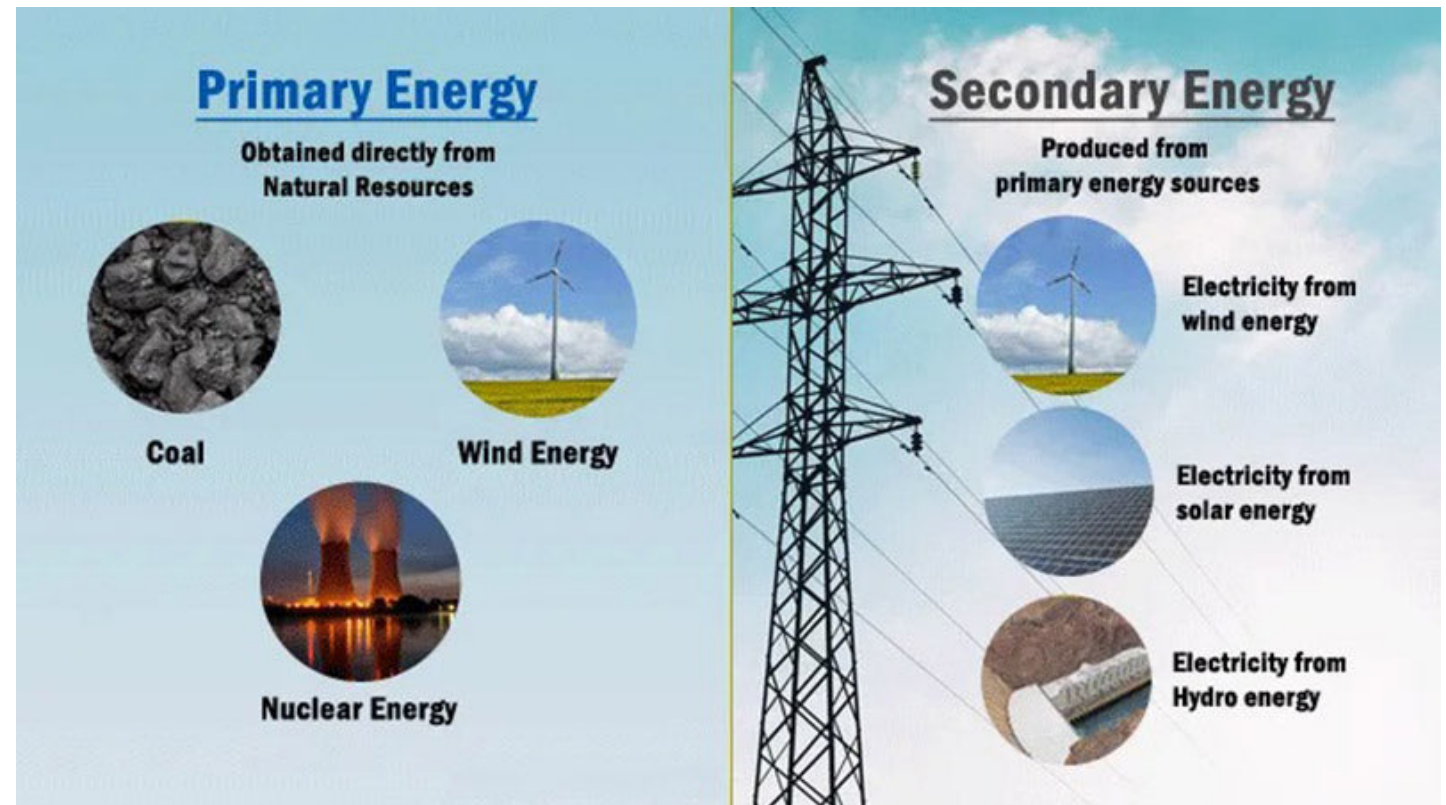
Title of Unit	Magnetic field
Title of Lesson	AC generator
Starter Activity	Revision Kahoot: induction
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- Demonstration: ac generator video - Inquiry: worksheet on various parts (name and purposes) of a generator - Question: Kahoot for induction revision
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	-
	International Mindedness:
	Invention of an generator takes the human society for a huge leap
Lesson Objectives	By the end of the lesson, students will be able to: • Describe a simple form of a.c. generator (rotating coil or rotating magnet) and the use of slip rings and brushes where needed. • Sketch and interpret graphs of e.m.f. against time for simple a.c. generators and relate the position of the generator coil to the peaks, troughs and zeros of the e.m.f.
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Inquirer
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirer, thinker
Teaching Activities	- Kahoot: induction - Worksheet: structure of an AC generator - Video: ac generator
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Exercise worksheet2 – force on current carrying wire

Generating electricity



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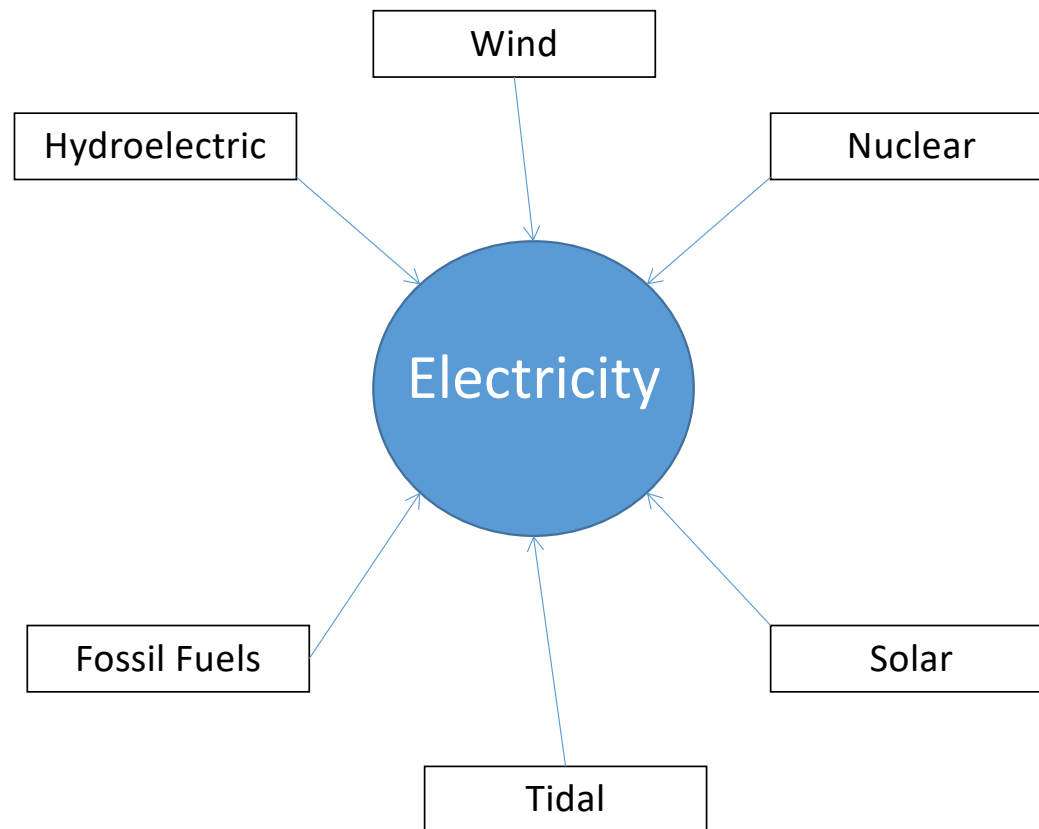
Electricity is known as a secondary energy source because it is produced using primary energy sources (e.g. Coal, nuclear fuel, wind power).



Generating electricity



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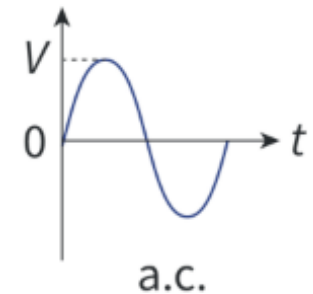
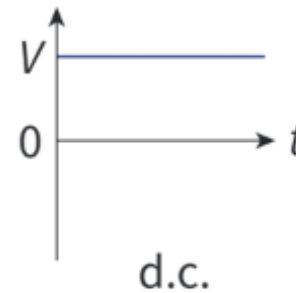
How is
electricity
generated?

Revision: a.c. and d.c. current



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- Current is the rate of flow of charges (electrons) around a circuit. We can generate two different types of current:
 - alternating current (a.c.): electrons continuously change direction.
 - direct current (d.c.): electrons flow in one direction only.
- A voltage source is needed to make both alternating current (a.c.) and direct current (d.c.).
 - for alternating current, the source must cause electrons to move back and forth
 - for direct current, electrons must flow in one direction.

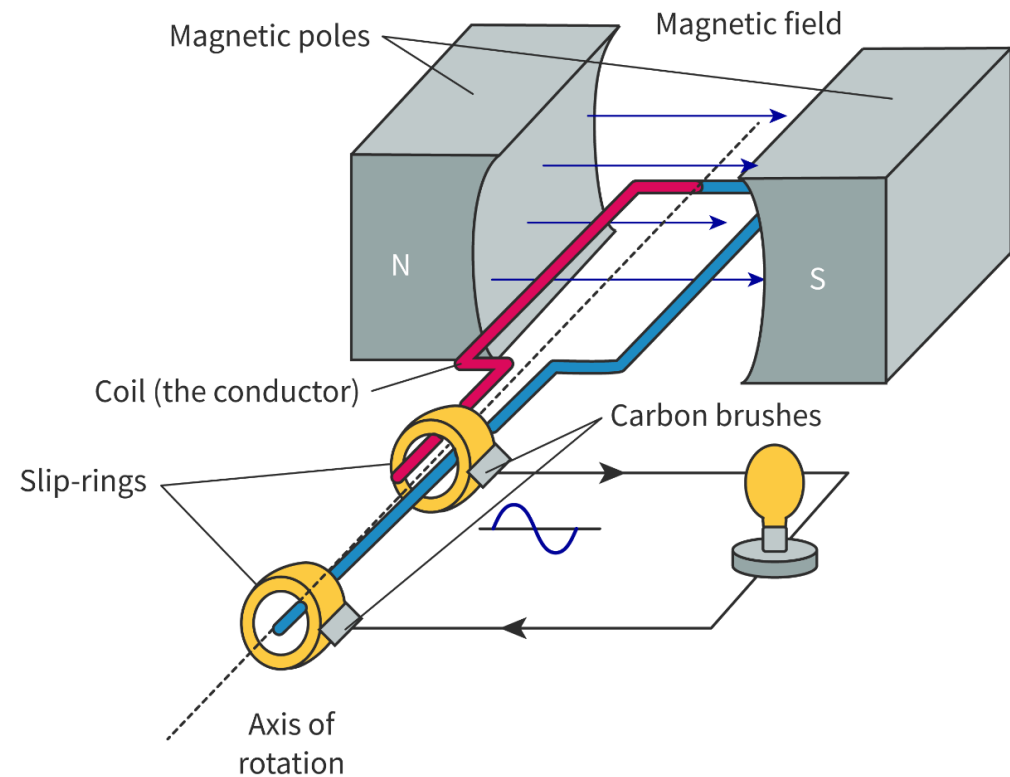


a.c. generator



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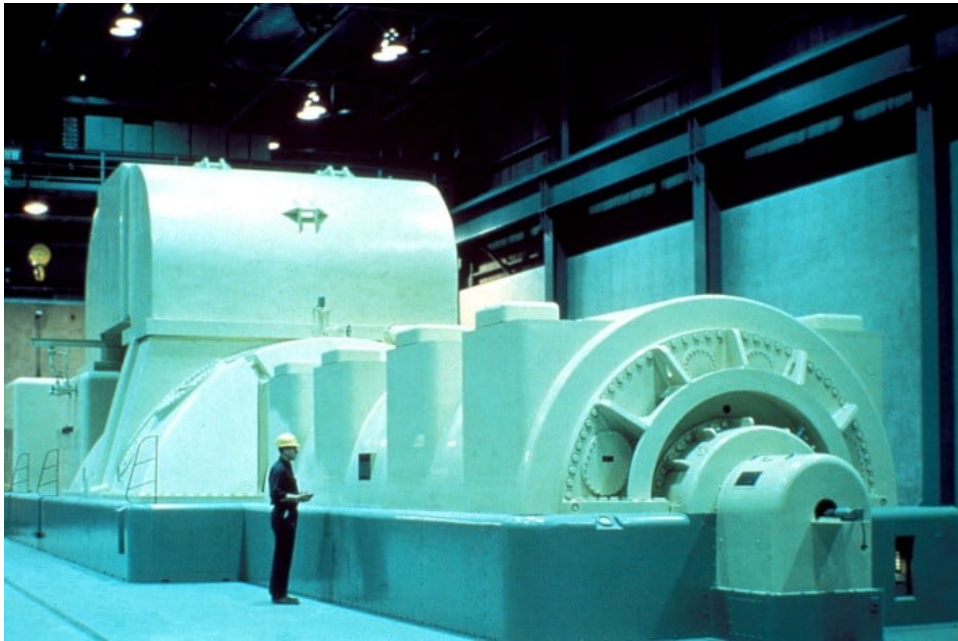
- A moving wire cutting magnetic field lines will induce an electromotive force (e.m.f.)
- The figure shows the arrangement of an A.C. generator, which allows a rectangular coil of wire to continuously rotate within a magnetic field, and so generate an induced e.m.f.



Practical a.c. generators



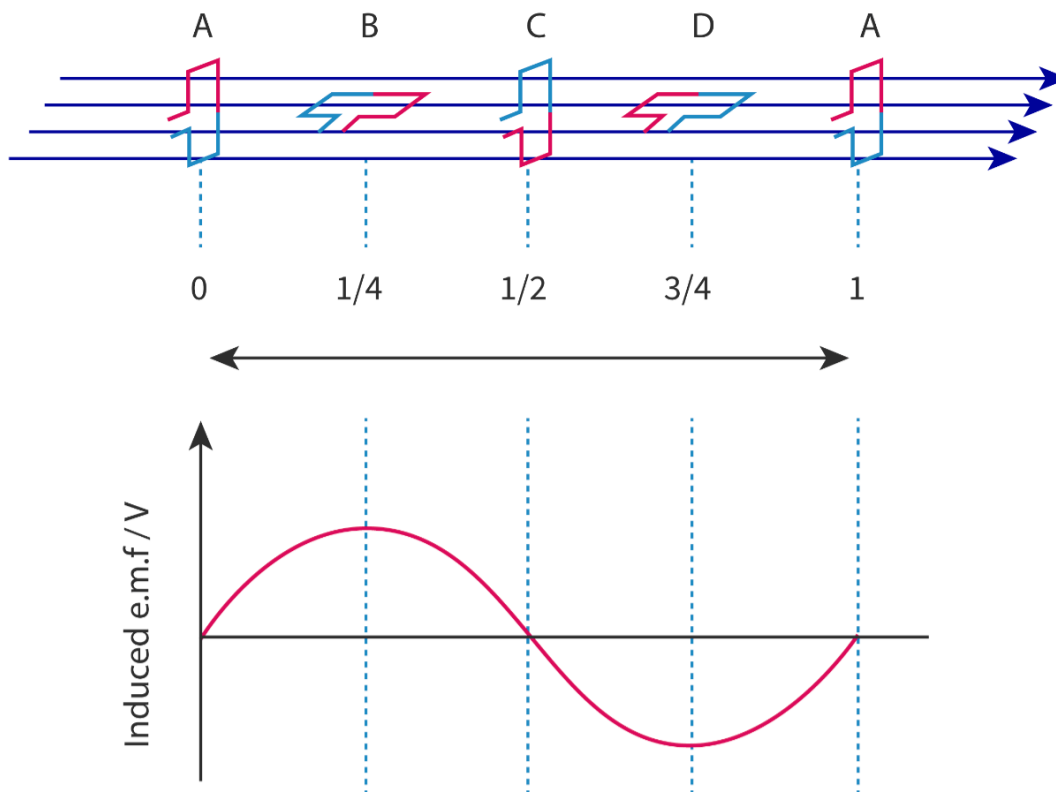
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Generating alternating current



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The graph below shows the direction and size of the induced e.m.f. during one complete rotation of the coil.

- **A** – The top and bottom sides of the coil are moving parallel to the magnetic field, and so no e.m.f. is induced.
- **B** – The long sides of the rectangular coil move exactly perpendicular (90°) to the magnetic field, and so maximum e.m.f. is induced.
- **C** – Again, the top and bottom sides of the coil are moving parallel to the magnetic field, and so no e.m.f. is induced.
- **D** – Again, the long sides of the coil move perpendicular (90°) to the field but in the opposite direction. Therefore, the induced e.m.f. is again maximum, but in the opposite direction.

Test your understanding



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Which of the following parts of a generator is responsible for allowing alternating current (a.c.) output?

1 Slip rings

2 Magnets

3 Coil

4 Armature

Slip rings

Explanation

The slip rings make constant contact with the rotating coil, so allowing the direction of the induced e.m.f. to alternate and create an alternating current.

Test your understanding



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At a certain point, the output of an alternating current (a.c.) generator is $+V$. What will the output be 180° later?

1 $-V$

2 $+V$

3 $+\frac{V}{2}$

4 0

$-V$

Explanation

After half a rotation, the output electromotive force (e.m.f.) will be equal in size but have the opposite direction.